

Name: _____ Tutorial group: _____

Matriculation number:

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September 2024 **MH1812 Test 1** 60 minutes

QUESTION 1. (30 marks)

Use mathematical induction to show that

$$1 + 2 \cdot 2 + 3 \cdot 2^2 + \cdots + n \cdot 2^{n-1} = (n - 1)2^n + 1.$$

for all integers $n \geq 1$.

[Proof:]

Base case: when $n = 1$, the left-hand side is 1 and the right-hand side is $0 \cdot 2^1 + 1 = 1$. The equality holds.

Inductive step: Suppose that the equality holds for n , and we shall show that it holds for $n + 1$, that is, we shall show that

$$1 + 2 \cdot 2 + \cdots + n \cdot 2^{n-1} + (n + 1) \cdot 2^n = n2^{n+1} + 1. \tag{1}$$

By induction hypothesis, the LHS of (1) equals

$$(n - 1)2^n + 1 + (n + 1)2^n = ((n - 1) + (n + 1))2^n + 1 = 2n \cdot 2^n + 1 = n2^{n+1} + 1,$$

which is exactly the RHS of (1).

Therefore, by mathematical induction, the original equality holds for all integers $n \geq 1$.

For graders only	Question	1	2(a)	2(b)	3(a)	3(b)	Bonus	Total
	Marks							

QUESTION 2. (30 marks)

(a) (10 marks) Find $5^{2024} \bmod 9$.

[Solution:] Observe that

$$\begin{aligned} 5^2 &\equiv 7 \pmod{9}, \\ 5^3 &\equiv 7 \cdot 5 \equiv -1 \pmod{9}, \\ 5^6 &\equiv 1 \pmod{9}. \end{aligned}$$

Since $2024 \bmod 6 = 2$, the answer is thus

$$5^2 \bmod 9 = 7.$$

(b) Let $\mathbb{Z}^+$ denote the set of positive integers. For $x, y \in \mathbb{Z}^+$, let $P(x, y)$ denote the predicate “ x/y is an even integer”. What are the truth values of these statements?

- (i) (10 marks) $\forall x \in \mathbb{Z}^+, \exists y \in \mathbb{Z}^+, P(x, y)$.
- (ii) (10 marks) $\exists x \in \mathbb{Z}^+, \forall y \in \mathbb{Z}^+, P(x, y)$.

[Solution:]

- (i) The statement is false. When $x = 1$, since $y \geq 1$, we always have $0 < x/y \leq 1$ and so x/y cannot be an even integer.
- (ii) The statement is false. For every $x \in \mathbb{Z}^+$, $x/y = 1$ is not an even integer for $y = x$.

QUESTION 3.

(40 marks)

(a) (20 marks) Determine whether $((p \wedge q) \rightarrow r) \rightarrow ((p \rightarrow q) \wedge (q \rightarrow r))$ is a tautology.

[Proof:] It is not a tautology. One can take $p \equiv T$, $q \equiv F$ and r to be any truth value.

(b) (20 marks) Determine whether the following argument is valid. (Please state the names of the inference rules whenever you apply them.¹)

$(p \rightarrow q) \wedge (p \rightarrow r);$
 $p \vee s;$
 $\neg q;$
 $\therefore s.$

[Solution:] The argument is valid, as shown by the following inference table.

Step	Formula	Reason
(1)	$(p \rightarrow q) \wedge (p \rightarrow r)$	Premise
(2)	$p \rightarrow q$	(1), conjunctive simplification
(3)	$\neg q$	Premise
(4)	$\neg p$	(2)+(3), modus tollens
(5)	$p \vee s$	Premise
(6)	s	(4)+(5), disjunctive syllogism

BONUS QUESTION.

(10 marks)

[Points will be given to *fully* correct solutions only. The total mark of this test is capped at 100 marks.]

Show that $\sqrt{n+2} + \sqrt{n-2}$ is irrational for all integers $n \geq 3$.

[Proof:] If $\sqrt{n+2} + \sqrt{n-2}$ is rational, then its square is also rational, thus $\sqrt{n^2-4}$ is rational. Suppose that $\sqrt{n^2-4} = p/q$ for some integers $p, q > 0$ such that $\gcd(p, q) = 1$. Then $n^2 - 4 = p^2/q^2$. Since the left-hand side is an integer, the right-hand side must also be an integer, it must hold that $q = 1$ and so $4 = n^2 - p^2 = (n+p)(n-p)$. Since $p > 0$, we must have $n+p = 4$ and $n-p = 1$, which has no integer solutions. Therefore, $\sqrt{n+2} + \sqrt{n-2}$ cannot be rational when $n \geq 3$.

[Note: the text in red is the most critical part of the proof. A common mistake is missing this line and jumping to that $n^2 - 4$ must be a perfect square.]

¹The inference rules you may use are: modus ponens, modus tollens, conjunctive simplification, conjunctive addition, disjunctive addition, disjunctive syllogism, hypothetical syllogism and rule of contradiction.